

Mod Rack

A 2U rack of four parallel effects - chorus, tremolo, vibe and delay - each with a circuit-bending MOD pot.



Overview

Mod Rack puts four classic modulation effects side by side in one rack unit: a bucket-brigade CHORUS, an optical TREMOLO, a photocell VIBE and a BBD / tape DELAY. They all work in parallel on the input, and each one adds only its own change to the sound: the dry signal always passes through, so turning a module up never takes anything away from the original.

Every module has a red MOD pot that bends its circuit in an unusual direction: the chorus's clock stretches into a warbly, seasick, lo-fi sound; the tremolo's smooth pulse chops into a square on / off; the vibe's lamp becomes lopsided and thumps; and the delay's clock runs away into self-oscillation with wow and flutter. Three L / C / R keys per module decide where its effect goes in the stereo field. Rates and times can follow the project tempo, and a gentle DRIFT keeps the LFOs from sounding mechanical.

Reach for it for

- Lush stereo chorus on guitars, keys and synths
- Surf and vintage tremolo, from smooth to choppy
- Throbbing, swirling vibe on guitar and electric piano
- Dub echoes and runaway, self-oscillating delay
- Stacking several modulation effects on one source in one plugin

Controls

Left: presets and size

LCD	Shows the current preset, or the name and value of the control you are touching. Click it to open the preset list.
PRESETS	Opens the list of factory presets. Choosing one sets all four modules, the master DRIFT and OUTPUT and the master MIX, and switches tempo sync off. The first entry, INIT, is the starting point with every module silent.
◀ / ▶	Step to the previous or next preset.
S / M / L	Panel size: small, medium or large (new instances open in Large). The panel also scales to fit the plugin window.
?	Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The four modules

From left to right: TREMOLO, CHORUS, VIBE and DELAY. Knobs: drag up and down (Shift for fine moves); double-click or Ctrl-click to reset.

TREMOLO (optical)

RATE	The tremolo speed, 0.1 to 15 Hz (4 Hz by default), or a note value when synced.
DEPTH	How deep the volume pulses, 0 to 100%. At 0 the module is silent.
SQUARE CHOP (MOD)	Morphs the smooth sine pulse into a hard square on / off chop, and makes the optical response faster to match.

CHORUS (BBD)

RATE	The chorus LFO speed, 0.05 to 8 Hz (0.8 Hz by default), or
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	a note value when synced.
DEPTH	How far the chorus sweeps, 0 to 100% (50% by default).
MIX	How much chorus is added, 0 to 100%. At 0 the module is silent.
CLOCK STRETCH (MOD)	Stretches the bucket-brigade line: longer delays, a lower clock (with its grainy aliasing and darker top), plus a random warble and a slow seasick wander. Turn it up for lo-fi, warped tape-like chorus.

VIBE (photocell)

RATE	The vibe speed, 0.1 to 12 Hz (1.5 Hz by default), or a note value when synced.
DEPTH	How deep the four-stage phase sweep goes, 0 to 100%. At 0 the module is silent.
ASYM LAMP (MOD)	Makes the lamp's sweep lopsided, slows its cooling, and adds a low-end thump in time with the pulse - the throbbing character of the classic vibe pedals.

DELAY (BBD / tape)

TIME	The delay time, 20 to 1500 ms (350 ms by default), or a note value when synced. Moving it glides the time, bending the pitch of the echoes like tape.
FEEDBACK	How many echoes, 0 to 95% (30% by default). The repeats get warmer and darker each time round, with a little saturation.
MIX	How much delay is added, 0 to 100%. At 0 the module is silent.
RUNAWAY CLOCK (MOD)	Pushes the feedback past the point of self-oscillation (the saturation keeps it in check) and adds wow and flutter to the time. For dub-style runaway echoes.

SYNC (RATE / TIME)

SYNC key or the RATE / TIME label

Click to step through: free (Hz or ms), SYNC (straight note values), DOTTED and TRIPLET. The label shows the mode and the knob then picks a note value from 4 BARS to 1/32. While REAPER plays, synced LFOs lock to the beat grid (drift does not move them). Synced delay times are limited to 1.5 s.

L / C / R

Three keys under each module choose where its effect goes. Click a key to switch it in or out.

L	Works on the left channel, out on the left.
R	Works on the right channel, out on the right.
C	Works on the mono sum, out in the centre (both sides).
L + R	A stereo effect (the chorus runs its two sides 90 degrees apart).
C with L and / or R	Spread: the voices run at different points in their cycle, or, for the delay, the taps sit at 3/4, 1 and 5/4 of the time.
All off	The module is muted (the LCD shows MUTED).

MASTER

DRIFT	A slow random wander of each module's LFO rate, independently, up to plus or minus 20% at 100% (20% by default). Synced LFOs are not affected while REAPER plays.
MIX	The master amount of all four modules together, 0 to 100% (100% by default). The dry signal is always at full level.
OUTPUT	Output level, -24 to +12 dB.
OUT meter	Output level.

LIMIT

Lights when the safety limiter is working. A limiter at the end holds the output just under 0 dBFS (about -0.3 dBFS).

Presets

Init	Every module silent: the starting point.
Lush Chorus	Slow, deep stereo chorus at 55%.
Seasick Tape	Slow, deep chorus with a lot of clock stretch on L, C and R, plus a 380 ms stereo delay with some runaway clock; more drift.
Surf Trem	Deep 6 Hz stereo tremolo with a little chop, and a short 120 ms slapback delay in the centre.
Vibe Throb	Deep vibe in the centre with a strongly lopsided lamp.
Dub Echo	450 ms delay on L, C and R with long feedback and some runaway clock.
Rotary Wash	Chorus, tremolo and a fast vibe together, for a swirling rotary-like wash.
Runaway	A 300 ms centre delay right on the edge of self-oscillation; output down 3 dB.

Tips

- Each module only adds its effect, so you can stack them freely: try a little chorus under the tremolo, or vibe into a short delay.
- Use the L / C / R keys to place effects: a tremolo on L and R only, with the chorus in C, makes a wide, moving image.
- RUNAWAY CLOCK with high FEEDBACK can build up - ride MIX or FEEDBACK with automation for dub throws.
- Lock the tremolo or delay to the tempo with SYNC so the pulse and echoes sit in the groove.

File: gfx-modrack.jsfx